Math-2205 Class Test 03 Section E

1. Consider $E[X] = -0.05$ to find unknown values from the following table. **[2]**

x	-2	-1	a	1	2
$P(X = x)$	0.2	0.25	0.1	b	0.15

2. Let a random experiment be the casting of a pair of fair *four-sided* dice and let X equal the minimum of two outcomes. With reasonable assumptions, find the variance of X. **[3]**

3. A continuous random variable X has the *cdf* as

$$F(x) = \begin{cases} l_1; & x < 5 \\ \frac{x-5}{5}; & 5 \leq x < 10 \\ l_2; & x \geq 10 \end{cases}$$

(a). What are the values of l_1 and l_2? **[1]**
(b). Estimate $P(7 \leq X < 12)$. **[1]**
(c). Find the mean and the median of the distribution. **[3]**

Math-2205 Class Test 03 Section G

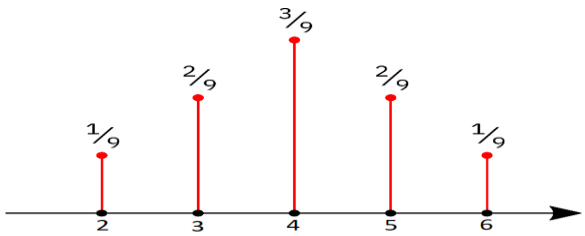
1. For $f(x) = \frac{x}{c}$; $x = 1, 2, \dots \dots 10$, determine the constant c for which $f(x)$ is the *pmf* for a random variable X. **[1]**

2. Let the random variable X have the *pmf* $f(x) = \frac{x}{10}$; $x = 1, 2, 3, 4$. Compute $P(X < 3)$, $E(2X^2 + 5)$, and $V(3)$. **[4]**

3. Let X have the *pdf* $f(x) = \frac{x^3}{4}$; $0 \leq x \leq 2$. Then, find and sketch the graph of *cdf*. Also, estimate the $P(X \geq 1)$ and median. **[5]**

Math-2205 Class Test 03 Section O

1. Line graph of a random variable X is given in the figure below. Compute $P(X \geq 3)$ and standard deviation. Also, find the median of X. **[5]**



2. Let X have the *pdf* $f(x) = x - kx^3; 0 \leq x \leq 2$. Find the value of k and then estimate the $P(X < 1)$, $P(X \leq 1)$, and mode of X. **[5]**

Math-2205 Class Test 03 Section P

1. In a bet, the betting person wins $1, $2 and $3 with probabilities 0.25, 0.2 and 0.15, and loses $1 with probability 0.4 for each $1 bet. Find $V(3-2X)$. **[3]**

2. Let a random experiment be the casting of a pair of fair *four-sided* dice and let X equal the difference of two outcomes. With reasonable assumptions construct the probability distribution table. **[2]**

3. Let $f(y)=\frac{3}{2}y^2\,;\,-1<y<1$ be the *pdf* of a continuous random variable Y. Find mean, $P(Y>0)$, and Q_1 of Y. **[5]**